

L6 :- Optimization - 2

Towards Gradient descent.

• Recap

- Differentiation geometric picture
- How to find derivative for any given functⁿ

Examples of finding derivatives of some functⁿ

- How to check if a functⁿ is differentiable

Checking differentiability of few functions

- Commonly used derivatives.

- Rules of differentiation

Linearity rule

product rule

quotient rule

~~and~~ chain rule

- How to use derivatives for optimization

How to identify critical points for any functⁿ.

How to identify if a critical point is minima or maxima

Recap

linear vector $\rightarrow$ distance, norm
dot product

unit vector

projected vector

weight vector $\perp$ to line

distance from origin / point
to line

$$= \frac{w_0}{\|w\|}$$

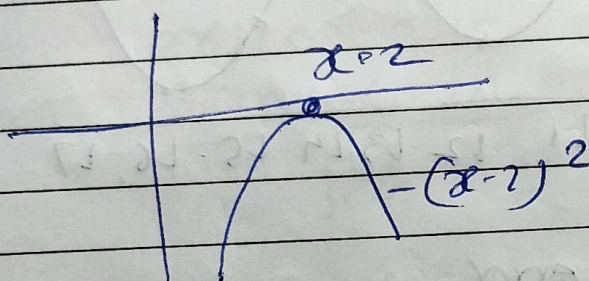
$$\frac{w^T x_1 + w_0}{\|w\|}$$

$$\text{Max } \sum \frac{w^T x_i + w_0}{\|w\|} (y_i)$$

Argmax

↳ max input value at which
o/p is max.

argmax at $-(x-2)^2$ is $x=0$
at which it is max i.e. 0



So $\underset{x}{\operatorname{argmax}} -(x-2)^2 = 0$

We want

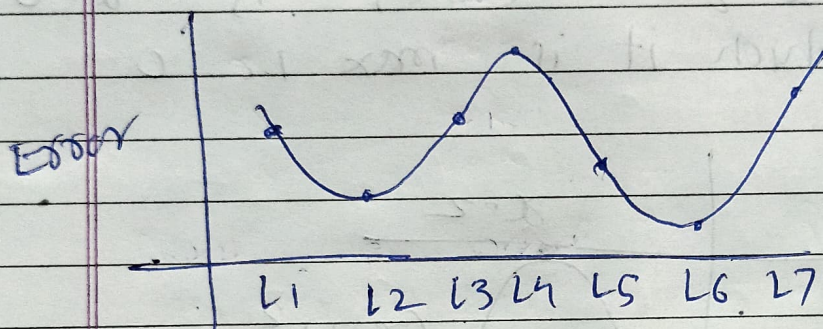
$$\underset{\omega, \omega_0}{\operatorname{argmax}} \sum_i \underbrace{\frac{\omega^T x_i}{\|\omega\|}}_{y_i} y_i$$

$$G(D, \omega, \omega')$$

$$\text{Gain} = -\text{Loss}$$

↓
max

↓
min

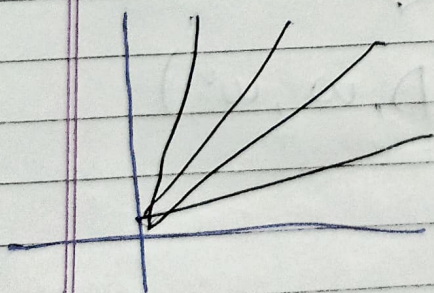


Min error

L_6 - min

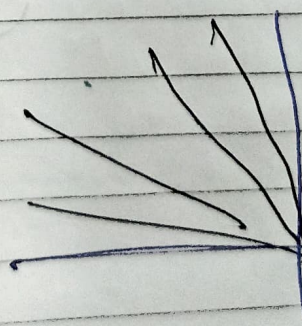
L_2 - low

slope at both point will be 0.
slope means change in x result
0 change in y



+ve slope

acute angle



-ve slope

obtuse angle

This is called as derivative
derivative is slope.

change in one value w.r.t.
change in another value.

e.g. in case of home
price

change in price

sqm $\leftarrow$ w.r.t.

in loss function

change in loss

$w, w_0 \leftarrow$ w.r.t.

take ex: of exp vs salary

exp. salary

1 10K

2 20K

3 30K

4 40K

5 50K

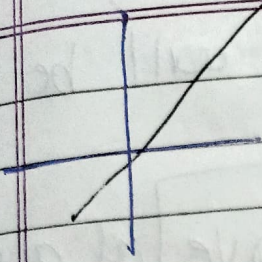
6 60K

Now we want to find w & w_0 for this, ignore w_0 for

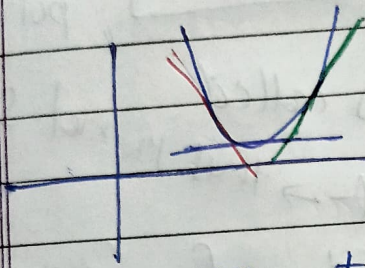
$$\text{salary} = w \cdot \text{exp}$$

$$\text{let } w = 5$$

for exp 6 it will 30
& we want the error
 $60 - 30 = 30$ to be minimum
by changing w .

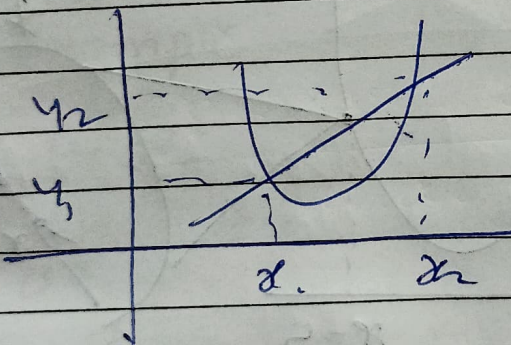


slope m is constant here



slope m is changing here

green +ve slope
red -ve slope
blue 0 slope

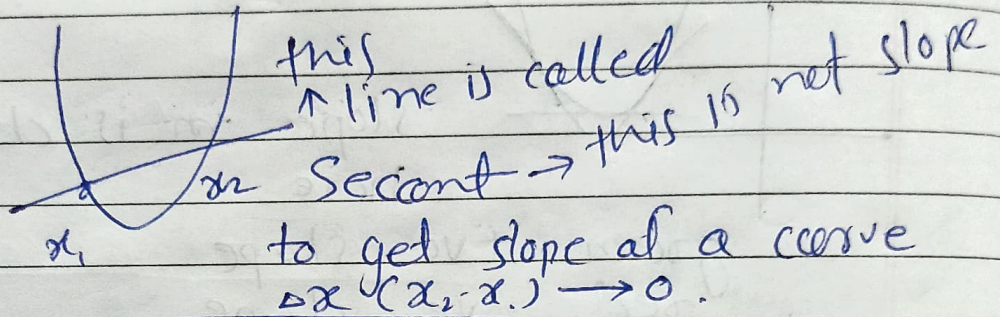


$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

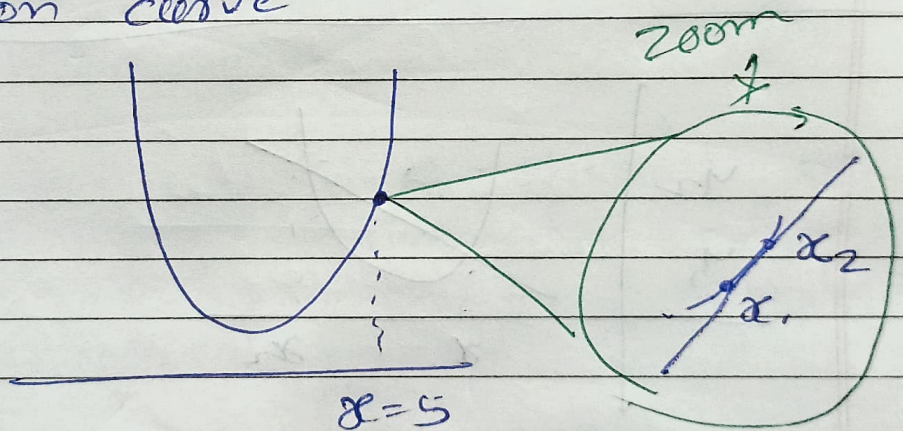
this will slope of line not curve

Here main issue will be finding

Slope of Curve at given point



finding slope at point x on curve



$$x_1 = 5$$

$$x_2 = 5.00001$$

$$x_2 - x_1 = \Delta x$$

$$\lim \rightarrow 0$$

$$m = \frac{f(x_2) - f(x_1)}{\Delta x}$$

$$m = \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

the above eqⁿ will become
derivative eqⁿ if $\Delta x \rightarrow 0$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

This will be slope of a
curve at x_1 ?

example

$$y = x^2$$

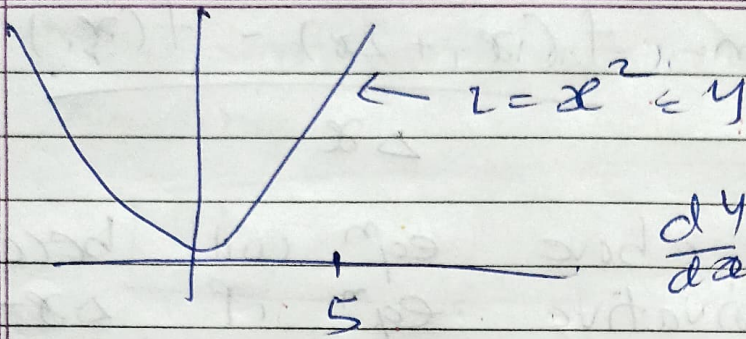
$$\frac{d}{dx} y = 2x$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x_1 + \Delta x) - f(x_1)}{\Delta x}$$

$$= \frac{(x + \Delta x)^2 - x^2}{\Delta x}$$

$$= \frac{x^2 + 2x\Delta x - x^2}{\Delta x}$$

$$= \frac{2x\Delta x}{\Delta x} = 2x$$



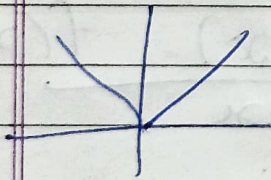
$$\frac{dy}{dx} = 2x$$

$$\frac{d}{dx} y \text{ at } x=5 = 2x = 10$$

$$\frac{d}{dx} y \text{ at } x=0 = 2x = 0$$

$$\frac{d}{dx} y \text{ at } x=-3 = 2x = -6$$

Let's find derivative of
 $y = |x|$



$$y = |x|$$

$$= \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$\frac{dy}{dx} = \frac{f(x+dx) - f(x)}{\lim_{dx \rightarrow 0} dx}$$

$$= \frac{f(x+dx) - f(x)}{dx}$$

$$= \frac{x + \Delta x - x}{\Delta x} \Rightarrow \text{for } (x > 0)$$

$$= 1$$

$$= \frac{-x - \Delta x + x}{\Delta x} \Rightarrow \text{for } (x < 0)$$

$$= -1$$

$$\text{LHD} = -1$$

$$\text{RHD} = +1$$

$$\text{LHD} \neq \text{RHD}$$

so this function is not differentiable

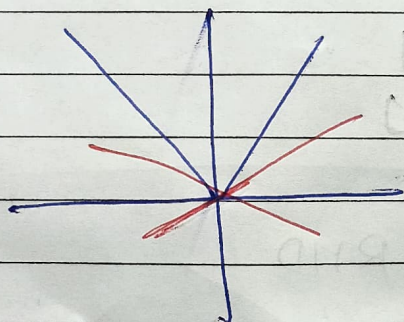
y is continuous

Is $|x|$ differentiable?

① check if it is continuous

② check if $LHD = RHD$

both ① & ② should be true to say a function is differentiable



$$y = |x|$$

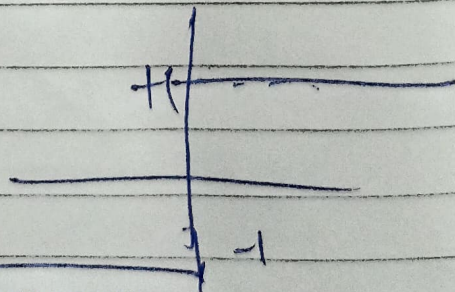
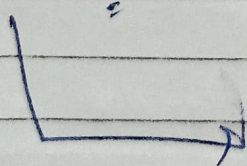
Multiple slope at $x=0$ hence not differentiable

$$y = |x|$$

$$\frac{dy}{dx} = 1$$

$$x > 0$$

$$x < 0$$



dy not continuous
hence not differentiable

Common Rules

$$(1) \frac{d}{dx} x^n = n x^{n-1}$$

$$(2) \frac{d}{dx} \log(x) = 1/x$$

$$(3) \frac{d}{dx} e^x = e^x$$

$$(4) \frac{d}{dx} \sin x = \cos(x)$$

$$(5) \frac{d}{dx} \cos x = -\sin x$$

$$(6) \frac{d}{dx} \tan x = \sec^2(x)$$

$$(7) \frac{d}{dx} c = 0$$

Rules

① Linearity

$$\frac{d}{dx} (f + g) = \frac{d}{dx} f + \frac{d}{dx} g$$

② Product rule

$$y = f(x) \cdot g(x)$$

$$\frac{d}{dx} y = \frac{d}{dx} f(x) \cdot g(x) +$$

$$f(x) \cdot \frac{d}{dx} g(x)$$

③ Quotient rule

$$y = \frac{f(x)}{g(x)}$$

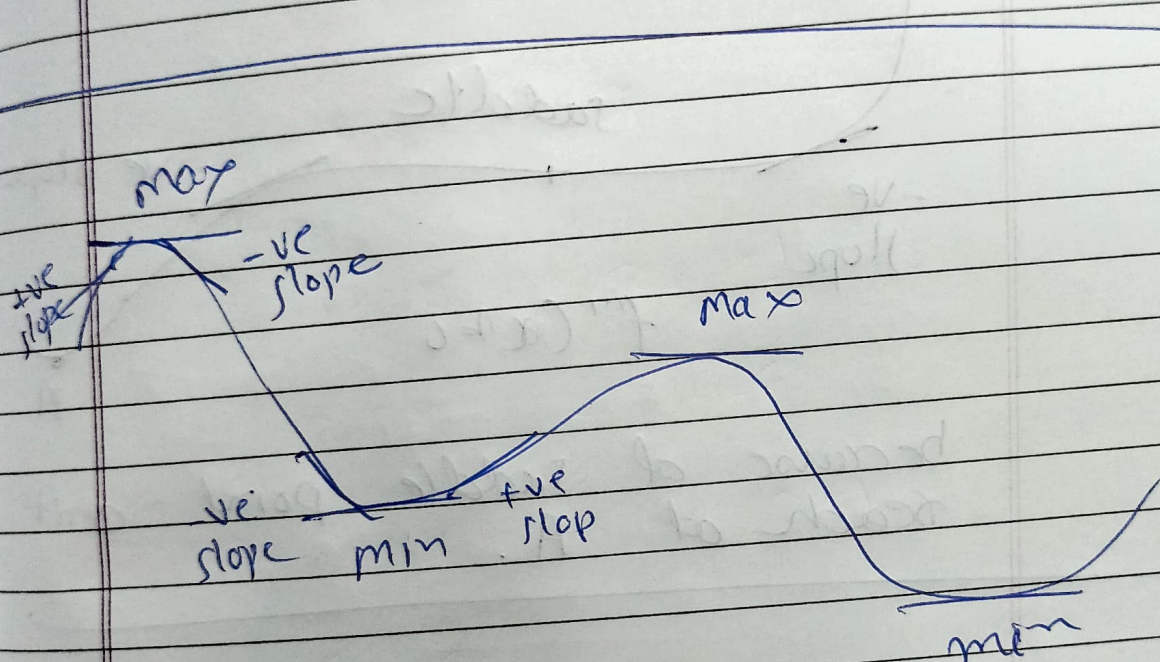
$$g(x) f'(x) - f(x) g'(x)$$

$$\frac{\quad}{(g(x))^2}$$

④ chain rule

$$y = f(g(x))$$

$$= f'(g(x)) \times g'(x)$$



max $\rightarrow$ slope change from
point +ve to -ve

min $\rightarrow$ slope change from
point -ve to +ve

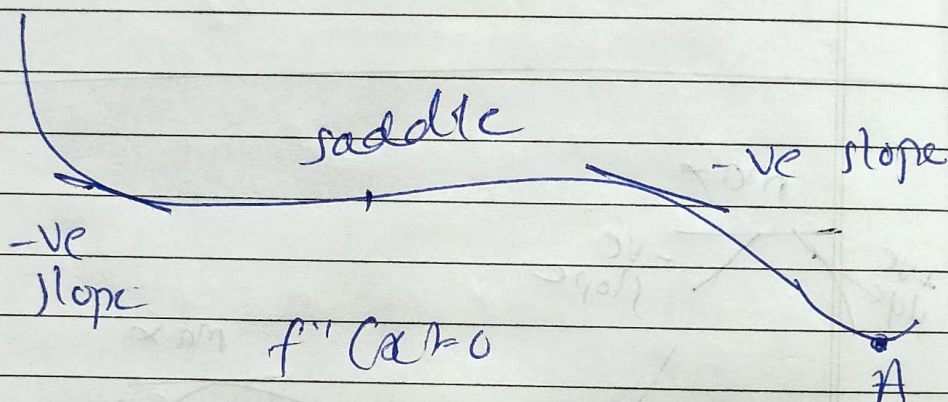
We are finding change in slope
which is double derivative
of curve

$$\frac{d}{dx} f(x) = \text{slope} = f'(x)$$

$$f''(x) \frac{d^2}{dx^2} f(x) = \text{change in slope}$$

if $f''(x) > 0 \Rightarrow \min$

$f''(x) < 0 \Rightarrow \max.$



because of saddle point crit
reach at A.

critical point $\rightarrow$ ~~the~~ max/min
point.